

GEOG 172

Spatial Inequality

An application of the Lorenz Curve, a non-spatial statistic, to the study of spatial inequality

Lecture 2-2

IMPORTANT IDEAS

- Standard statistics sometimes have spatial analogs that are useful as simple summaries of spatial patterns.
- The spatial center is an important idea, but it can be calculated several different ways
- We can adapt other statistics for use in a spatial setting

LORENZ CURVE

The Lorenz curve is the foundation of the Gini index, which some of you may have encountered in other settings. Its most common application is in the measurement of income inequality.

Here we will use it to examine equality in the spatial distribution of population

HOOVER INDEX

The largest vertical distance between the Lorenz curve and the 45-degree line. Interpreted as the proportion of people who would have to move from high-density areas to low-density areas, to equalize population density in all regions

$$H = \frac{\sum |x_j - \bar{x}|}{2n\bar{x}}$$

Hoover Indices and Hu Lines

From concept to computer

DEMONSTRATION

1. What distance to the coast maximizes inequality of the spatial distribution of the U.S. population?
2. Has that distance changed over time?
3. How well-identified is the selected distance?

TUTORIAL

There are many potential ways to identify The different regions of California. Your first task this quarter is to define the state's vague NorCal and SoCal regions, using different approaches that includes: mean center, median center, Hu line, and concepts from cognitive geography.

Expectations:

- We provide you with a fairly complete .Rmd to begin with
- Translate the statistics we learned in class and readings into the code
- Execute the code
- Answer the questions embedded in the notebook